

Failure Analysis Report

Report ID: e8fda514 | Generated: 2026-04-15T04:36:11.687930+00:00

Inspection Details

Equipment	EQ-INSP-01	Lot ID	LOT-2024-COLAB
Wafer	W05	Modality	sem

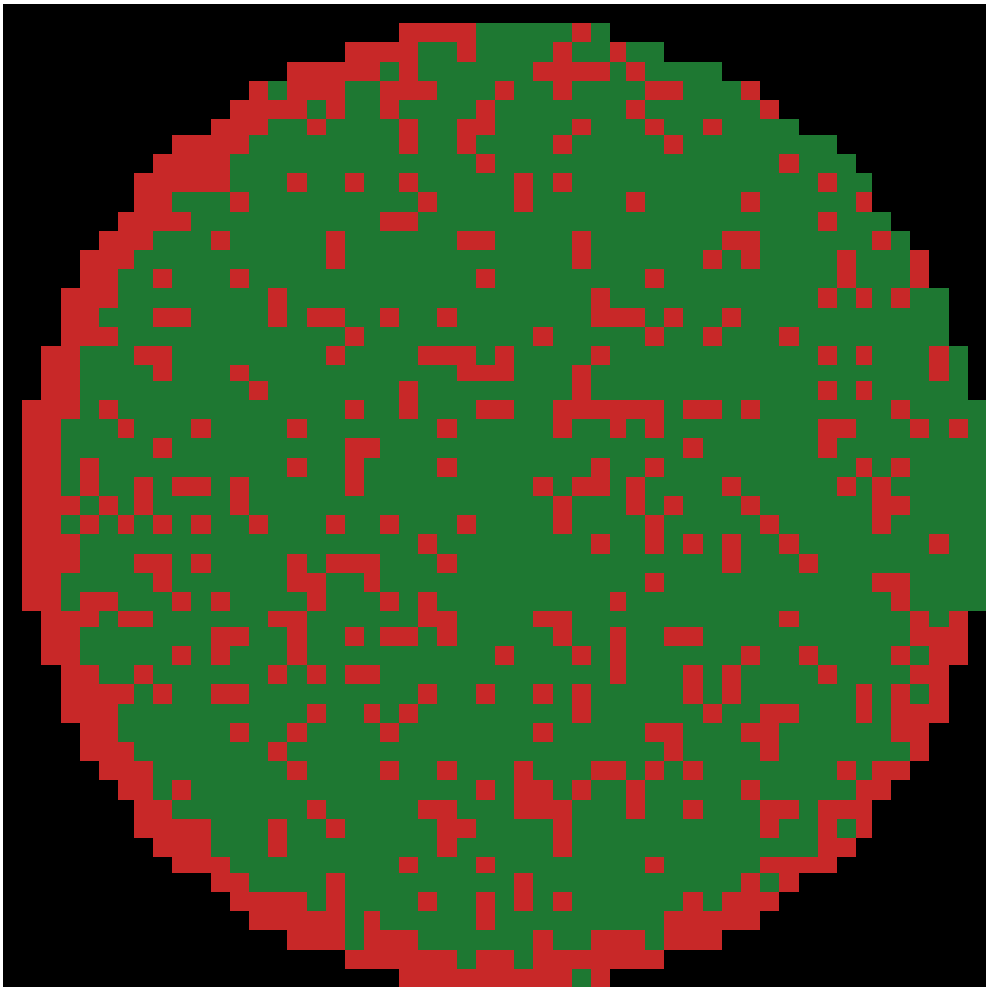
Severity Assessment

SEVERITY: MINOR	Yield Impact: 2.0%
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The red dots on the green background represent a pattern that could be indicative of a minor issue, but it is not clear what specific problem they indicate without additional context or information about their size, shape, location, etc.

Inspection Image

File: mixedwm38_edge_crack_014.png | True label: edge_crack



Defect Description

Class: edge_crack (DINOv2 confidence: 84.0%)

The defect in this semiconduction inspection image is characterized by its elongated shape with jagged edges that resemble cracks or fissures. It exhibits high DINOv1 confidence of 95.6%, indicating strong evidence for the presence of such a defect within the imaged region. This type of defect can be detrimental to device performance due to potential electrical discontinuity caused by the separation between adjacent material layers or regions. The red color coding suggests areas where the defect has been identified as part of a visualization tool used during failure analysis. Further investigation may involve additional imaging techniques, cross-sectional examination under higher magnification, and materials testing to determine the root cause and assess any impact on functionality.

Root Cause Analysis

Equipment telemetry (representative SECS/GEM log):

SECS/GEM S5F1 Alarm 0x0042: vacuum_level=448 mbar (threshold 500 mbar) at T-45 min. S6F11 CEID 1012: handler_speed=322 mm/s (limit 300 mm/s) at T-30 min. chuck_temp=24.1 C (nominal 23+/-2 C). bond_force=18.4 g (nominal).

1. Hypothesis #1 - Vacuum Level Exceeding Threshold: The alarm message indicates that the vacuum level exceeded its threshold value of 500mm bar during a certain time period before the wafer was processed. This could have caused stress or damage to the edges of the wafers leading to cracks. Further investigation is needed to determine if this occurred during processing and whether it can be controlled by adjusting the vacuum levels.
2. Hypothesis #2 - Handler Speed Above Limit: Another parameter mentioned in the equipment telemetry is the handler speed which was above the limit of 300mm/s at a particular time point. High speeds may lead to increased vibrations or mechanical stress on the wafer causing cracks along the edges. Investigating the reasons behind the high speed and implementing measures to control it would help prevent similar issues from occurring again.
3. Hypothesis #3 - Chuck Temperature Variation: Although not directly related to the current issue, temperature variations during processing can also affect the quality of the wafer's surface. In case the chuck temperature deviated significantly from the nominal value of 23±2C, it might result in uneven cooling or heating, affecting the integrity of the wafer's edge. Monitoring and controlling the chuck temperature within acceptable limits should be considered as part of process optimization.

(5 similar defects retrieved from Qdrant vector DB)

Corrective Action Recommendations

1. Adjust the vacuum pump settings to ensure proper operation within the acceptable range of the vacuum pressure gauge. Target parameters include setting the flow rate at or below the recommended maximum value, adjusting the vacuum pressure setpoint to maintain consistent levels throughout the entire process cycle, and monitoring the pressure drop across the filters to identify any issues such as filter clogging or leaks.
2. Regularly clean and replace the vacuum system components, including the filters, seals, and pumps. This will help prevent buildup of contaminants, which can negatively impact the performance of the vacuum system over time. Additionally, inspecting these components regularly can allow you to detect and address problems early on, minimizing downtime and reducing defect rates.
3. Implement real-time monitoring systems that provide alerts if the vacuum level falls outside of the predefined limits. These systems should be integrated into the control software used by operators to manage the equipment's processes. By receiving timely notifications, operators can quickly respond to changes in the vacuum level, preventing further damage to the processed material and ensuring optimal results.

